Lab08

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1. Preparing the data

```r
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
head(wisc.df)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
84358402         M       20.29        14.34         135.10    1297.0
843786           M       12.45        15.70          82.57     477.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
84358402         0.10030          0.13280         0.1980             0.10430
843786           0.12780          0.17000         0.1578             0.08089
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
84358402        0.1809                0.05883    0.7572     0.7813        5.438
843786          0.2087                0.07613    0.3345     0.8902        2.217
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
```

```
84348301   27.23      0.009110        0.07458      0.05661           0.01867
84358402   94.44      0.011490        0.02461      0.05688           0.01885
843786     27.19      0.007510        0.03345      0.03672           0.01137
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
84358402     0.01756             0.005115        22.54         16.67
843786       0.02165             0.005082        15.47         23.75
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
84358402          152.20     1575.0           0.1374            0.2050
843786            103.40      741.6           0.1791            0.5249
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
84358402          0.4000               0.1625         0.2364
843786            0.5355               0.1741         0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

Removing the Diagnosis from the data frame and creating a new vector:

```r
wisc.data <- wisc.df[ , -1]
diagnosis <- factor(wisc.df$diagnosis)
```

2. Exploratory Data Analysis:

```r
#finding number of observations
dim(wisc.data)
```

```
[1] 569  30
```

```r
#finding number of malignant diagnoses
table(diagnosis)
```

```
diagnosis
  B   M 
357 212 
```

```r
#using grep() to return which data in wisc.data has "_mean" in the column name
#and then counting that up using the length() function
length(grep("_mean", colnames(wisc.data)))
```

```
[1] 10
```

> **Q1: How many observations are in this dataset?**
>
> There are 569 observations in the dataset
>
> **Q2: How many of the observations have a malignant diagnosis?** There are 212 observations with a malignant diagnosis
>
> **Q3: How many variables/features in the data are suffixed with "_mean" ?**
>
> 10 variables are suffixed with "_mean" in the data.

3. PCA

Checking if the data need to be scaled before PCA:

```r
# to find the means:
summary(colMeans(wisc.data))
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
  0.00379   0.06808   0.21771  61.89071  15.73372 880.58313 
```

```r
#finding standard deviation:
summary(apply(wisc.data, 2, sd))
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
  0.00265   0.01925   0.07273  34.90472   4.10679 569.35699 
```

Seeing that the ranges of the means and s.d. are so wide, the data needs to be scaled.

Performing the PCA

```r
wisc.pr <- prcomp(wisc.data, scale = T)
summary(wisc.pr)$importance
```

```
                           PC1      PC2      PC3      PC4      PC5      PC6
Standard deviation     3.644394 2.385656 1.678675 1.407352 1.284029 1.098798
Proportion of Variance 0.442720 0.189710 0.093930 0.066020 0.054960 0.040250
Cumulative Proportion  0.442720 0.632430 0.726360 0.792390 0.847340 0.887590
                            PC7       PC8       PC9      PC10      PC11
Standard deviation     0.8217178 0.6903746 0.6456739 0.5921938 0.5421399
Proportion of Variance 0.0225100 0.0158900 0.0139000 0.0116900 0.0098000
Cumulative Proportion  0.9101000 0.9259800 0.9398800 0.9515700 0.9613700
                           PC12      PC13      PC14      PC15      PC16
Standard deviation     0.5110395 0.4912815 0.3962445 0.3068142 0.2826001
Proportion of Variance 0.0087100 0.0080500 0.0052300 0.0031400 0.0026600
Cumulative Proportion  0.9700700 0.9781200 0.9833500 0.9864900 0.9891500
                           PC17      PC18      PC19      PC20      PC21
Standard deviation     0.2437192 0.2293878 0.2224356 0.1765203 0.1731268
Proportion of Variance 0.0019800 0.0017500 0.0016500 0.0010400 0.0010000
Cumulative Proportion  0.9911300 0.9928800 0.9945300 0.9955700 0.9965700
                           PC22      PC23      PC24      PC25      PC26
Standard deviation     0.1656484 0.1560155 0.1343689 0.1244238 0.0904303
Proportion of Variance 0.0009100 0.0008100 0.0006000 0.0005200 0.0002700
Cumulative Proportion  0.9974900 0.9983000 0.9989000 0.9994200 0.9996900
                            PC27      PC28       PC29       PC30
Standard deviation     0.08306903 0.0398665 0.02736427 0.01153451
Proportion of Variance 0.00023000 0.0000500 0.00002000 0.00000000
Cumulative Proportion  0.99992000 0.9999700 1.00000000 1.00000000
```

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

From my results, 44.272% of variance is captured byt he frist principal component.

```
pca.var <- wisc.pr$sdev^2
pca.var.per <- round(pca.var/sum(pca.var)*100, 3)
pca.var.per
```

```
 [1] 44.272 18.971  9.393  6.602  5.496  4.025  2.251  1.589  1.390  1.169
[11]  0.980  0.871  0.805  0.523  0.314  0.266  0.198  0.175  0.165  0.104
[21]  0.100  0.091  0.081  0.060  0.052  0.027  0.023  0.005  0.002  0.000
```

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

We need 3 PCs to describe at least 70% of the variation in the data.

```
sum(pca.var.per[1:3])
```

```
[1] 72.636
```

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

We need 7 PCs to describe at least 90% of variance in the data.

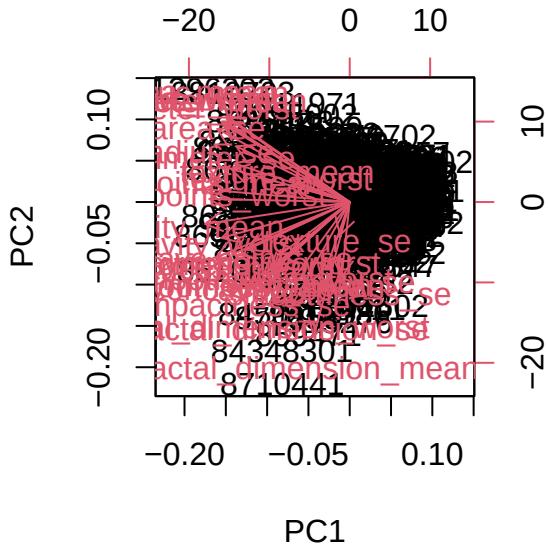
```
sum(pca.var.per[1:7])
```

```
[1] 91.01
```

Interpreting PCA results:

Creating a biplot:

```
biplot(wisc.pr)
```

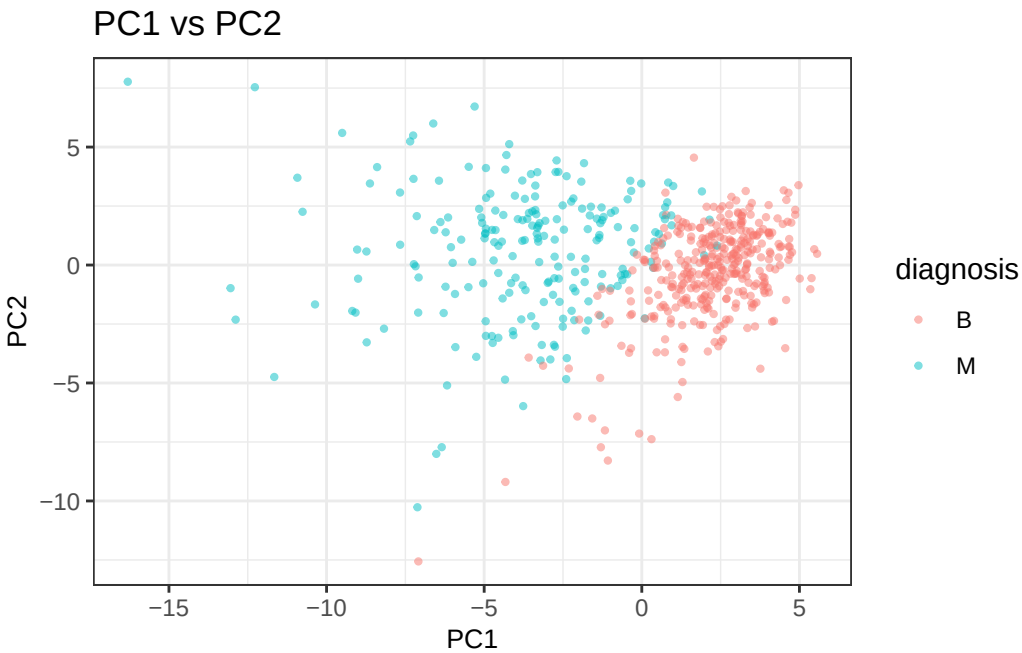


Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

The plot is really messy with a lot of points overlapping. It is very difficult to understand because it's trying to do a lot of things at once in an overlapping manner.

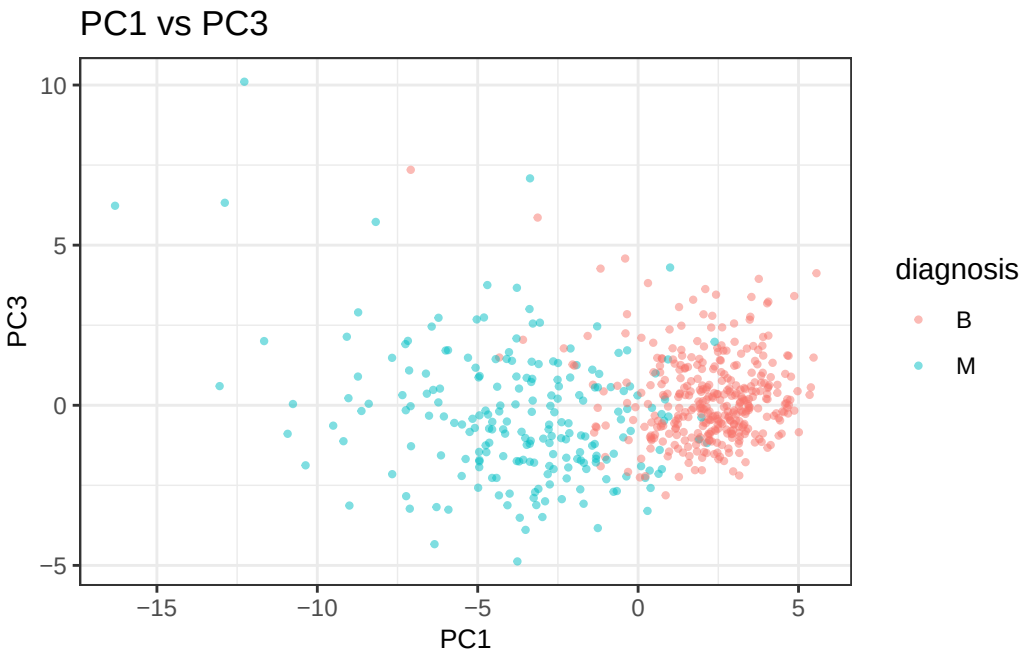
Using ggplot:

```
library(ggplot2)
ggplot(wisc.pr$x) +
  aes(x = PC1, y = PC2, color = diagnosis) +
  geom_point(size = 0.8, alpha = 0.5) +
  labs(title = "PC1 vs PC2", x = "PC1", y = "PC2") +
  theme_bw() +
  theme(
    axis.title = element_text(size = 10)
  )
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +
  aes(x = PC1, y = PC3, color = diagnosis) +
  geom_point(size = 0.8, alpha = 0.5) +
  labs(title = "PC1 vs PC3", x = "PC1", y = "PC3") +
  theme_bw() +
  theme(
    axis.title = element_text(size = 10)
  )
```



Variance explained

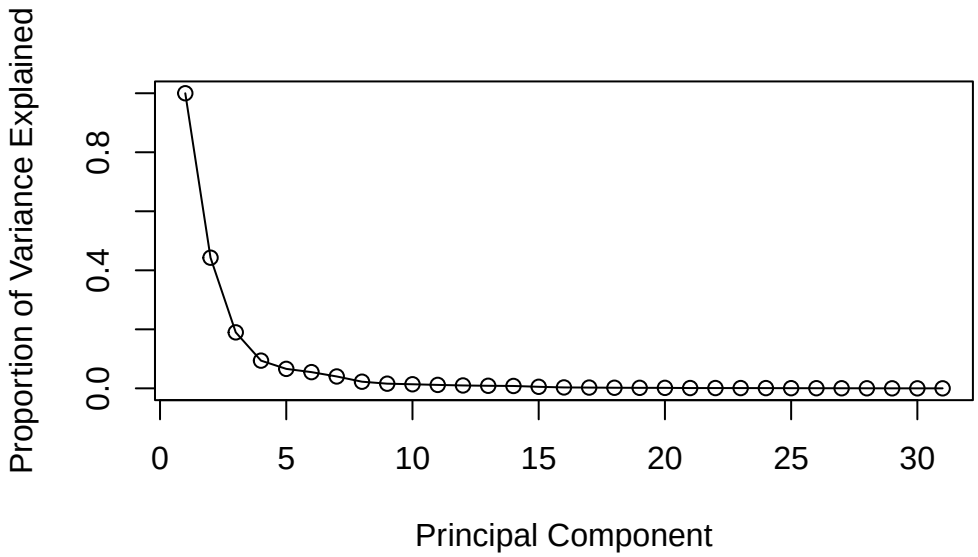
Calculating proportion of variance

```
# variance calculated above to answer questions 4 - 6
head(pca.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

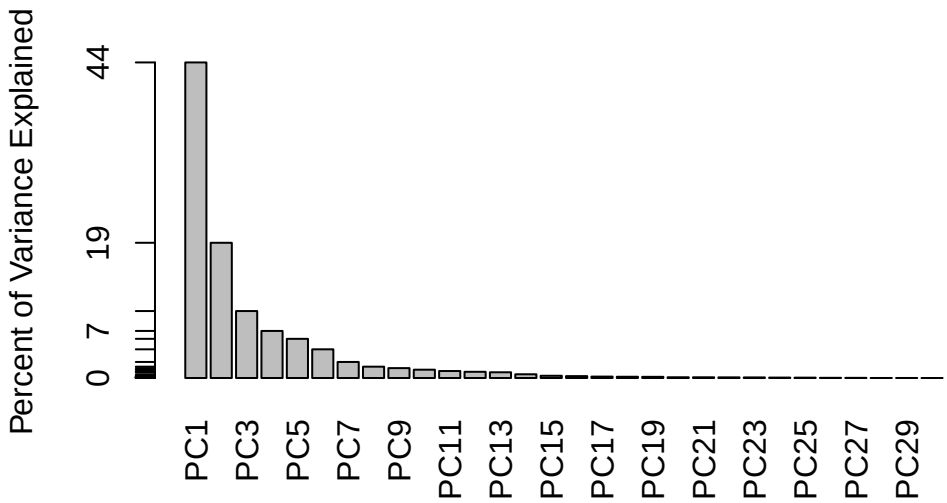
Plotting proportion of variance:

```
#calculating proportion of variance
pve <- pca.var/sum(pca.var)

#plotting this:
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



Alternate plot:

```
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```

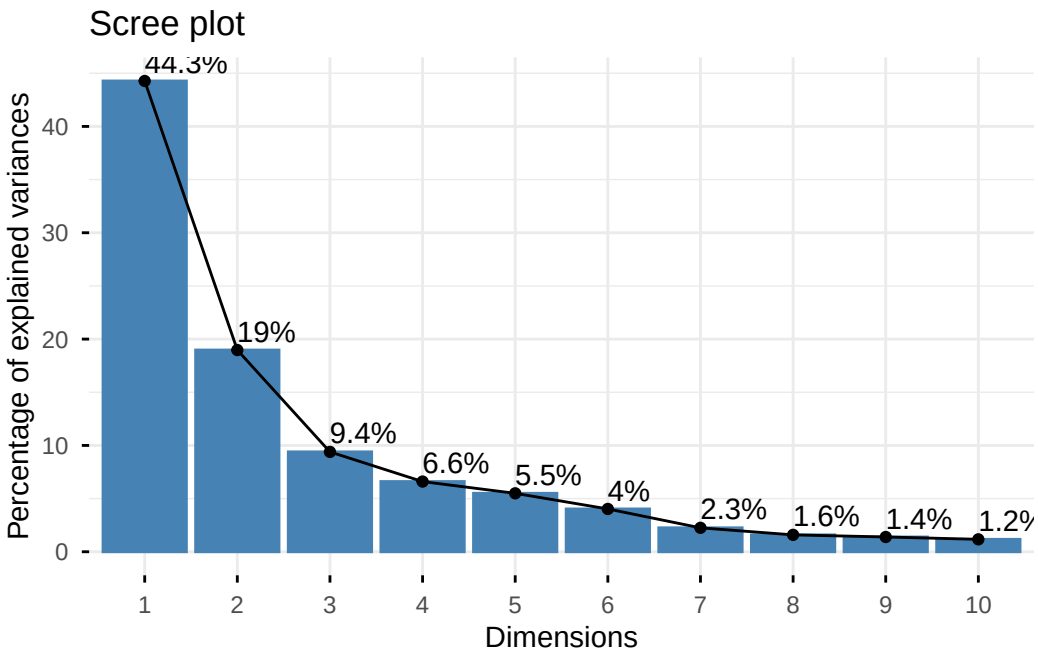


Using factoextra

```
library(factoextra)
```

```
Welcome to factoextra!

Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

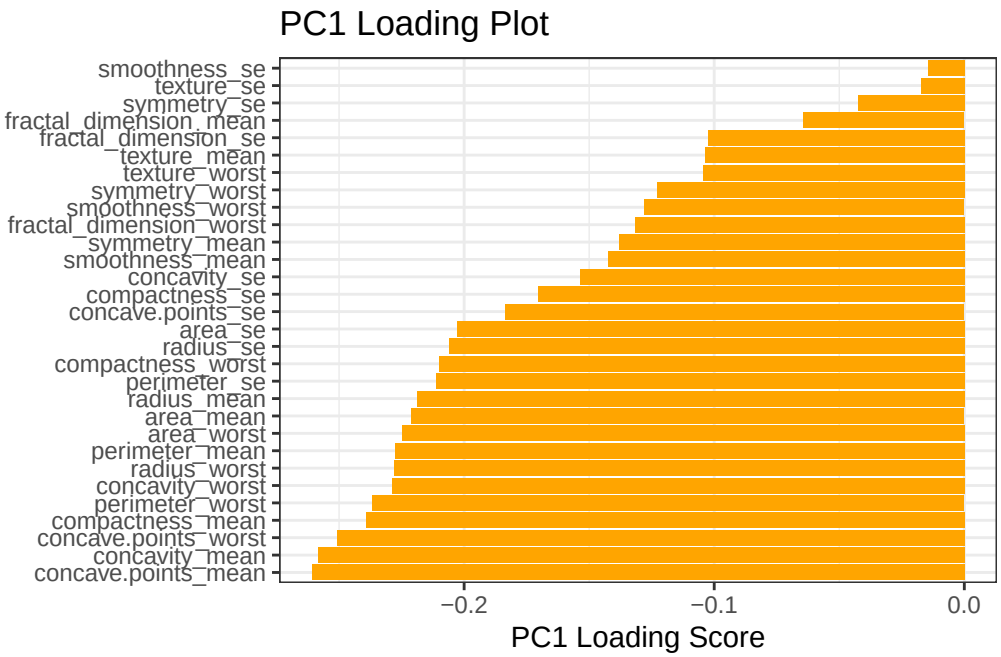
```r
fviz_eig(wisc.pr, addlabels = TRUE)
```



Communicating PCA results

Generating a loading plot for PC1:

```r
ggplot(wisc.pr$rotation) +
  aes(x = PC1, y = reorder(rownames(wisc.pr$rotation), PC1)) +
  geom_col(fill = "orange") +
  labs(title = "PC1 Loading Plot",
       x = "PC1 Loading Score",
       y = "") +
  theme_bw() +
  theme(axis.text.y = element_text(size = 9))
```



Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

The feature concave.points_mean contributes -0.2608538 to the mean. There are no features with larger contributions than this one, as seen by the plot above and also the ordered list below.

```
head(sort(wisc.pr$rotation[,1], decreasing = FALSE))
```

```
 concave.points_mean       concavity_mean concave.points_worst
          -0.2608538           -0.2584005           -0.2508860
    compactness_mean      perimeter_worst      concavity_worst
          -0.2392854           -0.2366397           -0.2287675
```

4. Hierarchical Clustering

Results of hierarchical clustering

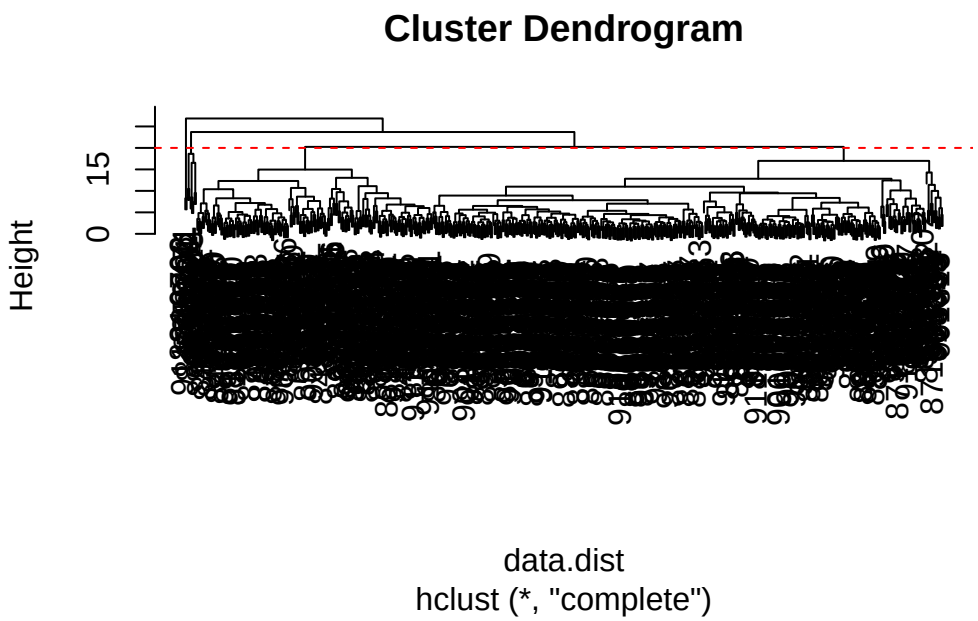
Scaling the data:

```r
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist, method = "complete")
```

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

At around y = 20, the model has 4 clusters.

```r
plot(wisc.hclust)
abline(h = 20, col = "red", lty = 2)
```



Selecting number of clusters

```r
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters)
```

```
wisc.hclust.clusters
  1   2   3   4
177   7 383   2
```

Using table()

```r
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

Q11. OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

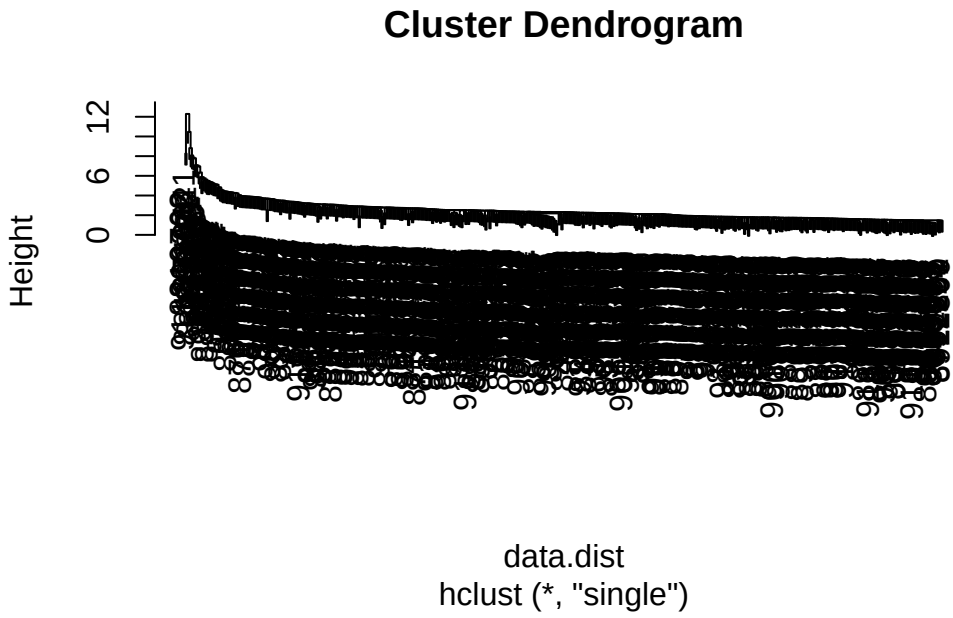
No, i could not find any number fo clusters between 2-6 that works better than k = 4 because anything k < 4 has almost equal malignant and benign cells in the same cluster and k > 4 has some groups with almost nothing in them.

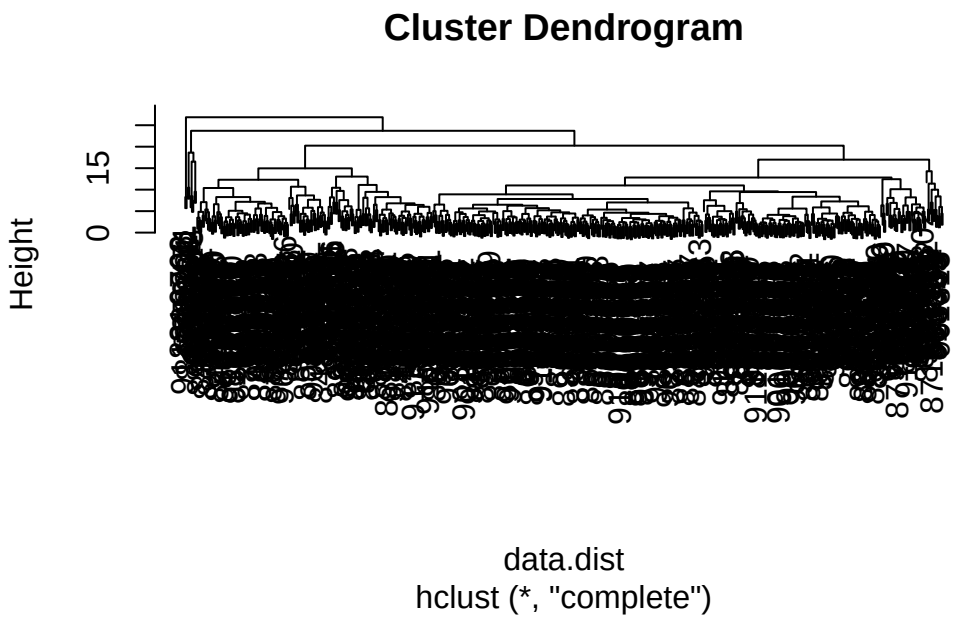
```r
repeat.clusters <- function(k) {
  table(cutree(wisc.hclust, k = k), diagnosis)}

sapply(c(2,3,5,6), repeat.clusters)
```

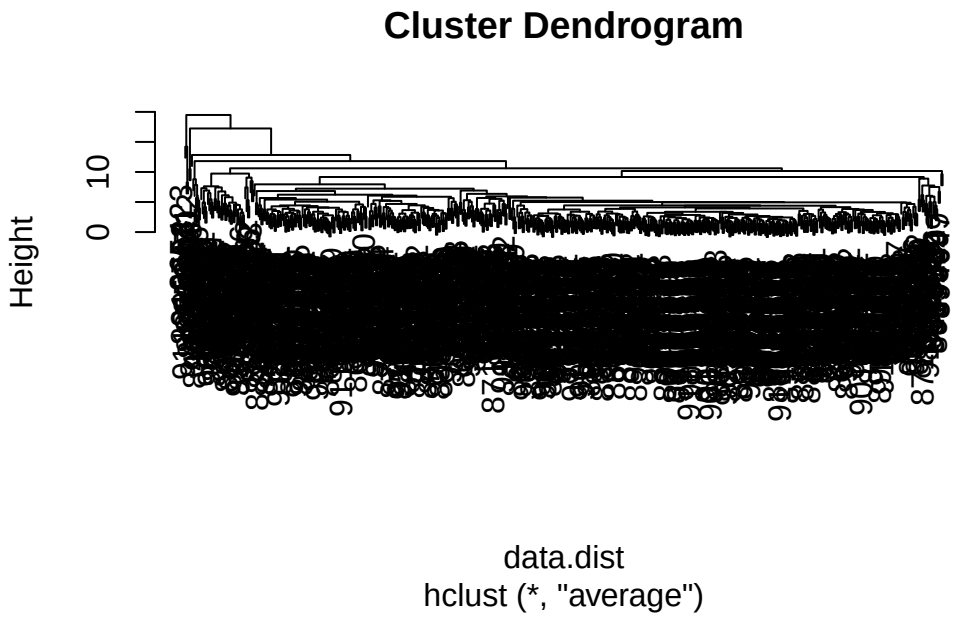
```
[[1]]
   diagnosis
      B   M
  1 357 210
  2   0   2

[[2]]
   diagnosis
      B   M
  1 355 205
  2   2   5
  3   0   2

[[3]]
   diagnosis
      B   M
  1  12 165
  2   0   5
```

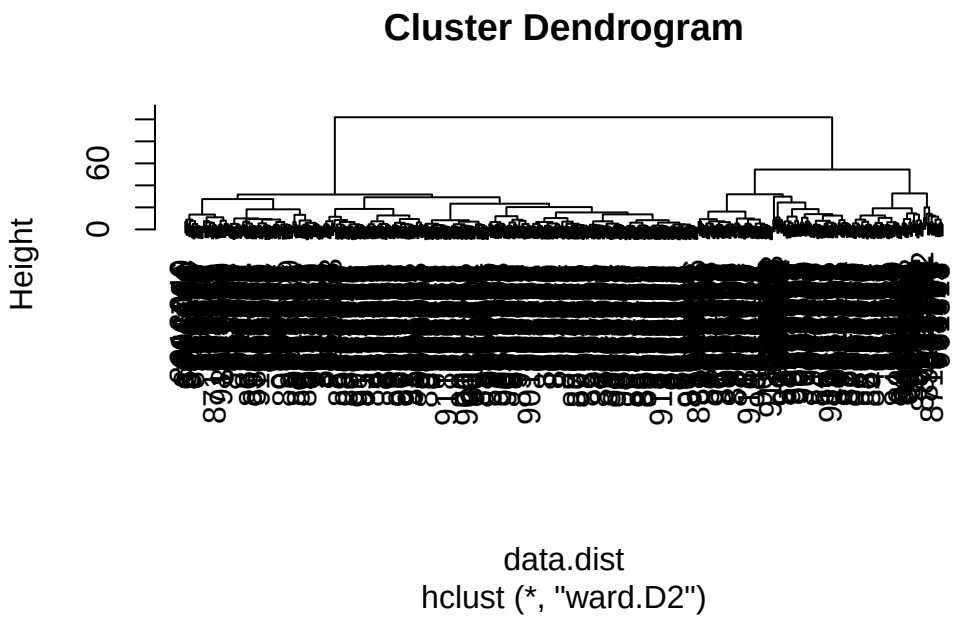
```
  3 343  40
  4   2   0
  5   0   2

[[4]]
   diagnosis
      B   M
  1  12 165
  2   0   5
  3 331  39
  4   2   0
  5  12   1
  6   0   2
```

```r
repeat.method <- function(x){
  plot(hclust(data.dist, method = x))}

sapply(c("single", "complete", "average", "ward.D2"), repeat.method)
```




Cluster Dendrogram
Height
0
15
data.dist
hclust (*, "complete")


Cluster Dendrogram
Height
0
10
data.dist
hclust (*, "average")




```
$single
NULL

$complete
NULL

$average
NULL

$ward.D2
NULL
```

```r
table(cutree(hclust(data.dist, method = "ward.D2"), k = 2), diagnosis)
```

```
   diagnosis
      B   M
  1  20 164
  2 337  48
```

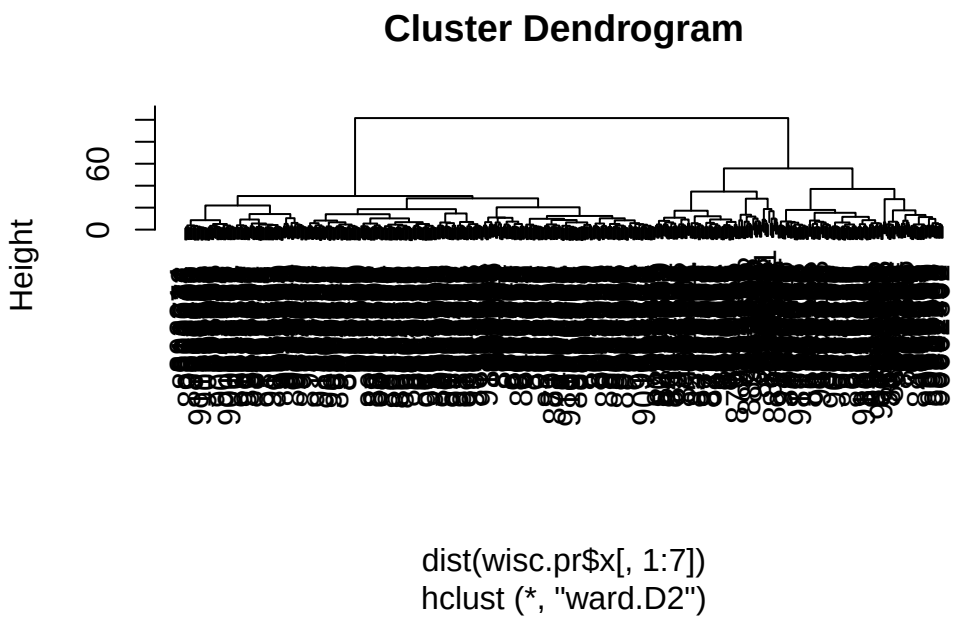
Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

The ward.D2 method gives me my favorite results for the same dataset because when i break the data in 2 clusters after applying this method fro hclust(), the resulting groups correspond very closely to the diagnosis of B or M.

5. Combining methods

Clustering on PCA results

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[ ,1:7]), method = "ward.D2")
plot(wisc.pr.hclust)
```



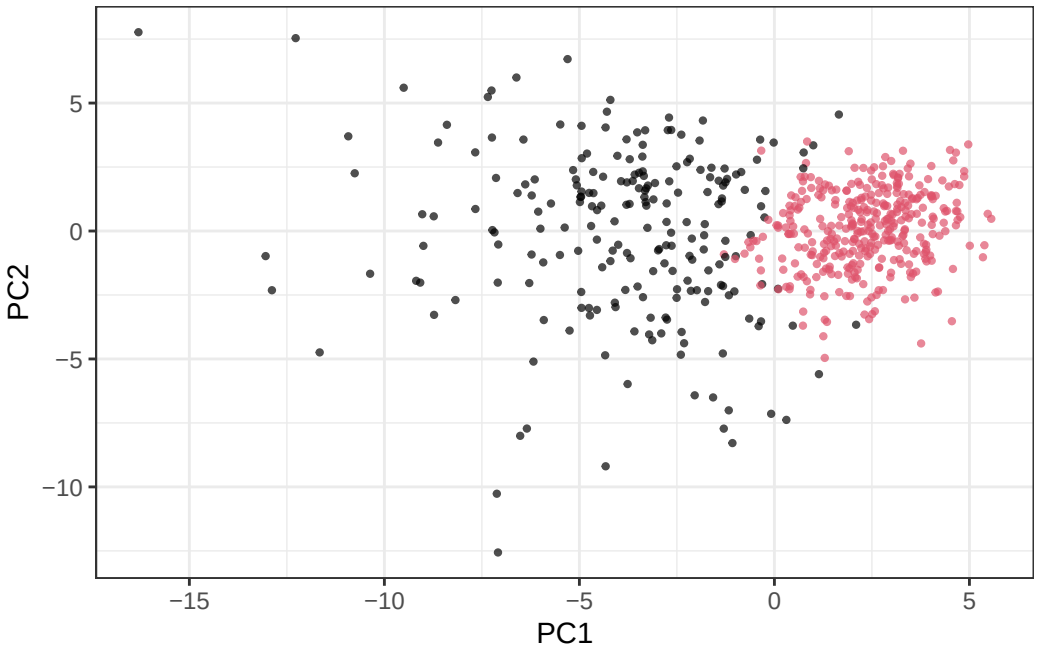
Visulaizing accuracy:

```
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

Plotting using ggplot:

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(color = grps, alpha = 0.7, size = 0.8) +
  theme_bw()
```



Optional: 3D plots using rgl

```
#library(rgl)
#plot3d(wisc.pr$x, xlab = "PC1", ylab = "PC2", zlab = "PC3", size = 1, col =
#grps, type = "s", cex = 1.5)
#rglwidget(width = 400, height = 400)
```

Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

The newly created model differentiates between M and B better since there are fewer clusters with more accurate groupings. In the previous hclust, there were 2 groups with almost no observations but fewer clusters resulted in inaccurate groupings. However, this hclust also shows slightly increased false positive and false negative readouts.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

The previous hclust models didn't do very well in differentiating the actual diagnoses since there was a larger number of groupings required in hclust than the actual diagnoses. It also included more false positives for malignant cells.

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

```
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

6. Specificity

Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

```
best.model <- table(grps, diagnosis)
sensitivity <- sum(best.model[ ,2])/sum(best.model[1,2], best.model[1,])
specificity <- sum(best.model[ ,1])/sum(best.model[1,1], best.model[2,])
best.model
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

7. Prediction:

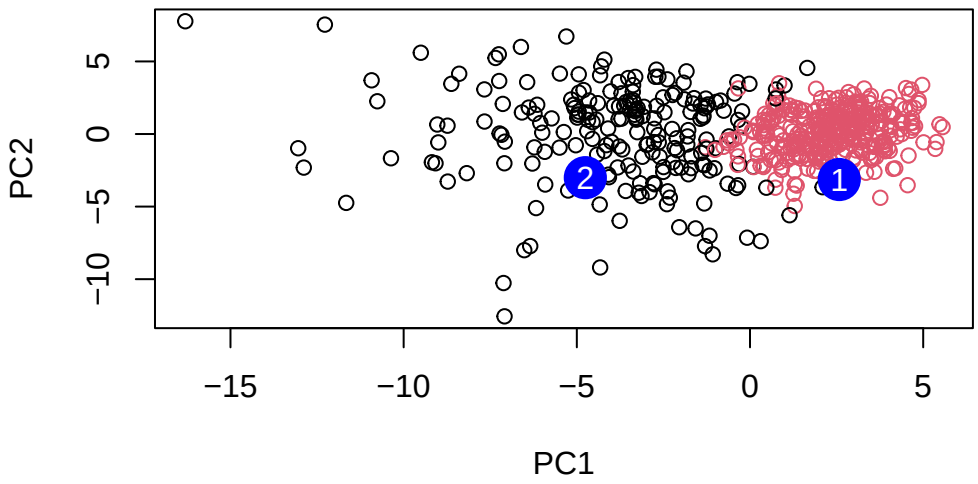
Using predict() to apply our PCA model to new data:

```
new.data <- read.csv("new_samples.csv")
npc <- predict(wisc.pr, newdata = new.data)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

Plotting the prediction of new data:

```
plot(wisc.pr$x[ , 1:2], col = grps)
points(npc[ ,1], npc[ ,2], col = "blue", pch = 16, cex = 3)
text(npc[,1], npc[,2], c(1,2), col = "white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

We should prioritize following up with Patient 2 since their cell nuclei measurements make them fall in the malignant category.